

BOB-066 Evidence “ Cross-Layer BOBA_UPSTREAM_PROXY Audit

Revision: 1 **Last modified:** 2026-08-15T12:15:00Z **Item:** BOB-066 “
Lava P3: BOBA_UPSTREAM_PROXY in download-proxy + qBitTorrent-go +
Jackett + compose env-forward **Status:** In progress (L1/L2/L4-partial
WIRED; L3 residual gap) **Verifier:** challenges/scripts/
upstream_proxy_wired_challenge.sh **Run id:** 20260815T115831Z

§11.4.108 Runtime signature “ three-mode verifier evidence

Anti-bluff coverage per §11.4.115 polarity switch + §11.4.107(10) self-validation. Each mode™s exit code is captured; every mode required to exit 0 for closure eligibility.

Mode 1 “ --self-validate (analyzer-honesty against a synthetic golden-BAD tree)

The verifier runs its OWN detector chain against a temp tree where every wire is DELIBERATELY ABSENT (stub main.py without apply_proxy_env, stub search.py with zero trust_env, stub httpx/proxy.go with no Configure/NewTransport/Proxy, empty docker-compose.yml). PASS = every detector correctly returned FAIL on the bad tree “ the analyzer is not a bluff.

=== self-validate: run detectors against a synthetic golden-BAD tree ===
PASS: self-validate “ every detector correctly FAILED on the golden-BAD

Exit: 0 “ analyzer honesty confirmed.

Mode 2 “ GREEN (RED_MODE=0, default) “ real tree

=== BOB-066 four-layer cross-layer verifier ===
RED_MODE=0 (0=GREEN guard, 1=reproduce pre-fix defect state)

--- L1: download-proxy (Python) ---

[PASS] L1 config/proxy.py exports apply_proxy_env + aiohttp_session_kwar
[PASS] L1 apply_proxy_env() invoked at boot in download-proxy/src/main.p
[PASS] L1 aiohttp ClientSession sites carry trust_env (9 uses of _tracke

--- L2: qBitTorrent-go ---

[PASS] L2 internal/httpx/proxy.go exports Configure + NewTransport + Pro
[PASS] L2 httpx.Configure(cfg.UpstreamProxy) called in cmd/qbittorrent-p
[PASS] L2 httpx.NewTransport() used in internal/api (torrent-download cl

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--- L3: Jackett ---
[SKIP] L3 Jackett proxy honor: reason=extension_absent (neither compose

--- L4: docker-compose.yml env-forward ---
[PASS] L4 docker-compose.yml service 'download-proxy' env-forwards proxy
[PASS] L4 docker-compose.yml service 'qbittorrent-proxy-go' env-forwards
[PASS] L4 docker-compose.yml service 'qbittorrent' env-forwards proxy va
[SKIP] L4 'jackett' env-forward absent (see L3 SKIP; tracked)
[SKIP] L4 'boba-jackett' env-forward absent (not tracker-bound: talks to
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Summary: PASS=9  FAIL=0  SKIP=3
Notes:
- L3 residual: Jackett has neither env-forward nor ServerConfig.ProxyUrl
=====
PASS: upstream_proxy_wired_challenge " every wired layer verified; resid

Exit: 0 " 9 PASS / 0 FAIL / 3 honest SKIP (Â§11.4.3 / Â§11.4.69
extension_absent).
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Mode 3 " RED (RED_MODE=1) " Â§11.4.115 polarity flip

Under RED each presence-detector's polarity flips: PRESENT code becomes FAIL (because the pre-fix defect state RED simulates would have it ABSENT). On a tree where the fix has landed, EXPECT many FAILs " that IS the signal the polarity switch + detectors work end-to-end (they would have caught the defect if the code were un-wired). fail == 0 under RED would mean detectors ignore polarity OR the tree was reverted " both bluffs surfaced here.

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Summary: PASS=0  FAIL=9  SKIP=3
=====
RED-MODE VERIFY: 9 presence-detectors correctly negated under RED " pola

Exit: 0 " 9 detectors correctly negated " RED"GREEN flip proven.
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Â§11.4.108 four-layer audit " per-layer runtime signatures

Layer Sub-requirement		Runtime signature	Status
L1	Python download-proxy honors	download-proxy/src/main.py:97 calls apply_proxy_env(); 9 aiohttp.ClientSession sites in	WIRED
	BOBA_UPSTREAM_PROXY (urllib subprocesss + aiohttp)	merge_service/search.py carry _tracker_session_kwargs()=trust_env; config/proxy.py exports the API + UPSTREAM_PROXY_ENV="BOBA_UPSTREAM_PROXY" sentinel	

Layer Sub-requirement		Runtime signature	Status
L2	qBittorrent-go http.Transport.Proxy honors BOBA_UPSTREAM_PROXY (or standard chain)	internal/httpx/proxy.go exports Configure/NewTransport/Proxy; cmd/ qbittorrent-proxy/main.go:29 calls httpx.Configure(cfg.UpstreamProxy) at boot; internal/api/download.go:65 installs httpx.NewTransport() on the torrent- download client; socks5:// supported natively (remote DNS)	WIRED
L3	Jackett "env-forward OR ServerConfig.ProxyUrl	Neither path present. jackett container has no proxy env-forward; no Go code touches ProxyUrl/ProxyType/ServerConfig in qBittorrent-go/internal/. download-proxy, qbittorrent-proxy-go, qbittorrent all env-forward	RESID GAP
L4	docker-compose.yml env-forward for every tracker-touching service	BOBA_UPSTREAM_PROXY + HTTP_PROXY + HTTPS_PROXY + NO_PROXY (with default bypass hosts). jackett + boba-jackett do NOT (boba-jackett is loopback-only so not required).	3/5 WIRED (Jackett gap = 1)

Recommended follow-up for L3

File a scoped ticket (e.g. B0B-066b) implementing ONE of: - **(a)** Add proxy env-forward to the jackett service block in docker-compose.yml (mirror the block at lines 15-20 of the qbittorrent service). Simplest fix. - **(b)** Add qBittorrent-go/internal/jackettconfig/proxy.go that on Jackett-config load reads BOBA_UPSTREAM_PROXY and PATCHes Jackett's ServerConfig.json ProxyType/ProxyUrl fields (Jackett will honor them for its own indexer HTTP egress). Requires a boba-jackett boot-time step + safe file-write discipline.

Status decision (Â§11.4.6 / Â§11.4.108 / Â§11.4.226)

BOB-066 is NOT closed as Completed because L3 remains un-wired: closing with 3/4 sub-requirements met would be a Â§11.4.226 evidence-class-at-closure violation (the reported defect layer includes L3, and a source-green closure of an un-wired layer is exactly the pattern the anchor forbids). Bumped Queued â†’ In progress; item_history entry #65 records the audit with this evidence path; description carries the AUDIT NOTE 2026-08-15 block naming L1/L2/L4 wired + L3 residual + follow-up recommendation.